

Not All Policy Shocks Are Alike: A Cohort Event-Study Atlas of the 2022–2025 U.S. Shock Wave

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Abstract

The years 2022–2025 delivered an unusually dense sequence of *dated* U.S. policy and technology shocks—from the CHIPS Act and ChatGPT to Silicon Valley Bank, the 2025 reciprocal tariffs, and the Reg-NMS half-cent tick. We assemble them into a single, uniformly measured event-study atlas across about 2,000 large firms and ask how the shocks differ along three axes: how much the market *moves*, how much it *differentiates* firms, and how it *propagates*. Three findings survive formal testing. First, cross-sectional differentiation is *real*: for the tariff shock, the Reg-NMS tick, and the SVB collapse, the event-window dispersion of firm CARs exceeds a placebo

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distribution of pseudo-event dispersions (ratios 1.5–1.9, empirical $p < 0.05$), and shock rankings by dispersion are stable across windows and dispersion measures (Spearman 0.72–0.83). Second, this differentiation is overwhelmingly *idiosyncratic*: market beta explains at most 14% of the cross-sectional CAR variance and usually far less, so the dispersion is firm-specific repricing—exactly what a firm-level exposure measure should capture. A pooled firm×shock regression on observable traits (size, beta, idiosyncratic volatility) explains 21% of |CAR|; the rest is the space reserved for the cohort’s hand-collected exposure test. Third, we are honest about what the small cross-section of shocks cannot support: the widely quoted claim that differentiation “scales with shock size” rests on a correlation of 0.69 whose bootstrap 95% interval is $[-0.04, 0.97]$ and which falls to 0.26 when the tariff shock is dropped—with $n = 10$ shocks the design can only reliably detect correlations above 0.78, so we report this as suggestive, not established. Propagation is heterogeneous ($I^2 = 99\%$ across shocks; SVB the negative-drift outlier), but a high-powered pooled test detects modest post-event *continuation* (within-shock, firm-clustered top-minus-bottom announcement-quintile drift +0.73%, $t = 4.0$). The atlas is a reusable scaffold; its completion is each student’s exposure measure.

JEL: G14, G18, E44. **Keywords:** event study, market efficiency, policy shocks, cross-sectional dispersion, randomization inference, meta-analysis.

1 Introduction

Empirical finance studies shocks one at a time. Yet 2022–2025 handed markets a whole *sequence* of clean, dated shocks, and studying them together—on identical footing—asks a question no single event study can: *how do policy shocks differ?* This cohort capstone builds that comparison. Using the shared market-model machinery behind the individual papers in this program (Fama, 1970; MacKinlay, 1997; Brown and Warner, 1985; Campbell et al., 1997), we compute cumulative abnormal returns (CARs) for a common sample of about 2,000 large firms around ten dated shocks (Table 1) and characterize each along three axes: the aggregate *level* move, the cross-sectional *differentiation* of firms, and the *propagation* of the reaction over the following days.

The first axis is the level move—how much the market index shifted. These range from a +3.1% pop on ChatGPT to a −11.0% two-day crash on the April-2025 reciprocal tariffs. The second, and the heart of the paper, is *differentiation*: how much firms diverged, measured by the cross-sectional standard deviation (SD) of announcement CARs. Because a market-wide shock is absorbed by the benchmark, the mean abnormal return is near zero by construction; the informative object is the *dispersion*, which measures how sharply a shock sorted winners from losers (Roll, 1988; Morck et al., 2000). The third axis is *propagation*: does the reaction complete quickly, drift, or reverse (Bernard and Thomas, 1989; De Bondt and Thaler, 1985)?

A descriptive atlas is only a starting point; the contribution of this draft is to subject each axis to a formal, design-based test, and to be candid about which claims the ten-shock cross-section can and cannot bear. We add ten tests to the original three-table draft.

Differentiation is real, and it is idiosyncratic. A shock that genuinely sorts firms should produce cross-sectional dispersion larger than what the same firms show on ordinary days. We build a placebo engine: one market model per firm over the full sample, then the cross-sectional SD of two-day CARs at the true event date compared against a distribution of pseudo-event dates drawn away from all ten shocks. For the tariff shock, the Reg-NMS half-cent tick, and the SVB collapse, the event dispersion is 1.5–1.9 times the placebo mean

and lands in the top few percent of the placebo distribution ($p < 0.05$; Table 4, Figure 2). The ranking of shocks by dispersion is not an artifact of the window or the statistic: it is stable across the $[0, +1]$, $[0, +5]$, and $[-5, +10]$ windows and across SD, interquartile range, and median absolute deviation (Spearman 0.72–0.83, all $p < 0.02$; Table 5). Crucially, the differentiation is not a beta story: a firm-by-firm regression of announcement CARs on market beta explains at most 14% of the cross-sectional variance (for the tariff) and essentially nothing for most shocks, leaving 92–100% of the dispersion idiosyncratic (Table 6). This is the empirical signature that firm-specific *exposure*, not systematic risk, drives the sorting—which is precisely the variable the cohort reserves for hand-collection.

Observable traits explain part of it; exposure is the reserved next step.

Pooling all 19,256 firm $\times$ shock observations, we regress $|\text{CAR}[0, +1]|$ on observable firm characteristics—log market capitalization, market beta, and idiosyncratic volatility—with shock and industry fixed effects and standard errors clustered by firm (Table 7). All three enter positively and idiosyncratic volatility dominates ($t = 11.3$); together with the fixed effects they explain 21% of the variation in reaction magnitude. The remaining variation is the space the design deliberately leaves open: whether each shock’s *specific* firm-level exposure (import-input intensity for tariffs, uninsured-deposit ratios for SVB, AI-exposure for the language-model shocks) explains the residual dispersion, and whether that exposure was priced *before* the announcement. That test requires the exposure crosswalk the cohort is assembling, and we do not attempt it with the generic observables here.

We are honest about the small cross-section. The draft’s headline was that differentiation “scales with shock size,” summarized by a 0.69 correlation between the absolute market move and the cross-sectional dispersion. With only ten shocks this is fragile. A bootstrap gives a 95% interval of $[-0.04, 0.97]$; leave-one-shock-out drops the correlation to 0.26 when the tariff shock is removed (Table 12, Figure 4); and a power calculation shows that with $n = 10$ the design can only reliably detect correlations above 0.78 (Table 13). The Reg-NMS tick makes the point vivid: it differentiates firms as sharply as the tariff (disper-

sion 5.1%) despite a tiny -1.1% market move. We therefore *retract the strong scaling claim* and report it as suggestive. What is robust is that differentiation is real, idiosyncratic, and largest for the tariff, the tick, and SVB.

Propagation is heterogeneous but shows modest continuation. Post-event drift differs enormously across shocks: a random-effects meta-analysis (DerSimonian and Laird, 1986; Higgins and Thompson, 2002) of the mean drift returns a pooled estimate indistinguishable from zero but with $I^2 = 99\%$ (Table 8, Figure 3)—these are genuinely different events, and SVB is the negative-drift (-3.5%) contagion outlier. Yet pooling firms and sorting by announcement CAR, the highest-power test we run, reveals a monotone pattern: firms that fell most on impact keep falling and firms that rose keep rising, a top-minus-bottom quintile drift of $+0.73\%$ ($t = 3.9$; Table 9)—modest *under-reaction*, not the over-reaction a contrarian would predict.

We make four contributions. First, we provide a *comparative* event-study lens on a dense natural sequence of policy shocks, measured on strictly common footing. Second, we show how to make a cross-shock atlas rigorous with design-based tools—placebo/permutation dispersion tests, rank-stability checks, pooled clustered regressions, random-effects meta-analysis, bootstrap, and minimum-detectable-effect calculations—rather than leaning on a handful of fragile cross-shock correlations. Third, we correct an over-strong claim in the original draft and quantify exactly why $n = 10$ cannot support it. Fourth, we locate the differentiation in idiosyncratic, firm-specific repricing, setting up the cohort’s reserved exposure test as the natural completion. Section 2 describes the shock wave and hypotheses, Section 3 the literature, Section 4 the data and design, Section 5 tests whether the differentiation is real, Section 6 decomposes it into systematic and idiosyncratic parts, Section 7 confronts the fragility of the size-scaling claim, Section 8 studies propagation, Section 9 reports power, and Section 10 concludes with the reserved exposure test.

2 The Shock Wave and Hypotheses

Ten dated shocks. We study ten dated events spanning industrial policy, tax, technology, banking, disclosure, trade, crypto regulation, and market microstructure (Table 1): the CHIPS Act signing (Aug 2022), the IRA buyback excise tax (Aug 2022), the ChatGPT release (Nov 2022) and GPT-4 release (Mar 2023), the Silicon Valley Bank collapse (Mar 2023), the SEC five-day 13D rule (Feb 2024), the April-2025 reciprocal tariff shock and the tariff pause days later, the GENIUS Act (Jul 2025), and the Reg-NMS half-cent tick pilot (Nov 2025). Each is “owned” by the cohort member whose individual project studies it; the ownership is what makes the reserved exposure test feasible. We use a single calendar date per shock and are explicit (Section 11) that the exact intraday timing—announcement versus effective, pre- versus post-close—is part of the reserved hand-collection and is not resolved here.

Three axes and hypotheses. The atlas measures each shock on three axes. The *level* is the raw market move over $[0, +1]$. *Differentiation* is the cross-sectional SD of firm CARs over $[0, +1]$: a large, uniform shock moves every firm the same way (low dispersion), while a shock that reallocates value across firms produces high dispersion. *Propagation* is captured by the mean drift over $[+2, +10]$ and by the cross-firm correlation between the announcement CAR and the subsequent drift (positive = continuation, negative = reversal). These motivate three hypotheses:

H1 (Real differentiation). Large policy shocks produce cross-sectional CAR dispersion above the level firms exhibit on ordinary days.

H2 (Idiosyncratic differentiation). The dispersion is firm-specific, not a repricing of systematic (beta) risk.

H3 (Efficient propagation). Post-event drift is small and mean-reverting, consistent with prompt price discovery, absent a contagion mechanism.

Our evidence supports H1 for the three largest shocks, supports H2 broadly, and gives a

nuanced verdict on H3: propagation is efficient on average but with a systematic component of modest under-reaction and one clear contagion outlier (SVB).

The ten shocks, one by one. Because each shock is owned by a cohort member, it is worth walking the atlas across its three axes (all figures from Table 1). The *CHIPS Act* signing (owned by A. Tang) moved the market +1.7% with modest dispersion (2.9%) and a small negative drift. The *IRA buyback excise tax* was nearly a non-event in aggregate (−0.7%) with the lowest dispersion of all (2.5%). The two language-model shocks bracket the technology story: *ChatGPT* (N. Ngo) produced a +3.1% pop and 3.1% dispersion, while *GPT-4* moved the index only +1.0% yet still dispersed firms (3.2%). The *SVB collapse* (A. Zhang) is the propagation outlier: a −3.7% move, 4.2% dispersion, and a large −3.5% drift with a positive reversal correlation (+0.42)—firms that fell most kept falling, the contagion signature. The *SEC five-day 13D rule* was procedurally invisible (−0.1%, 2.8% dispersion). The *April-2025 reciprocal tariff* (D. Yaveroglu) is the largest shock on every axis: −11.0% market move and 5.8% dispersion, its +6.2% *pause* days later partly reversing it with a strong positive drift (+2.6%). The *GENIUS Act* (D. Jórabekova) barely moved the broad index (+0.1%) as a targeted crypto-regulatory event. Finally, the *Reg-NMS half-cent tick* pilot (K. Borra) is the atlas’s most instructive case: a tiny −1.1% aggregate move but the second-highest dispersion (5.2%)—a microstructure rule that reshuffled firms without moving the market, the sharpest counterexample to “differentiation scales with size.”

3 Related Literature

This paper connects four strands. The first is *event-study methodology*. The market-model CAR and its inference are standard (Brown and Warner, 1985; MacKinlay, 1997; Campbell et al., 1997), but short-horizon tests are vulnerable to event-induced variance and to cross-sectional correlation of abnormal returns (Boehmer et al., 1991; Kolar and Pynnönen, 2010). A market-wide policy shock is the extreme case: abnormal returns are strongly

cross-sectionally correlated and the event mechanically inflates variance. This is exactly why we do not test mean CARs (which are near zero by construction here) but instead test the *dispersion* against a placebo distribution and pool firm $\times$ shock reactions with firm-clustered standard errors—design-based analogues of the corrections these papers propose.

The second strand is the *cross-sectional dispersion of returns* as information. Roll (1988) showed that firm-specific return variation is large and largely unexplained by systematic factors, and Morck et al. (2000) linked cross-firm return synchronicity to how much firm-specific information is capitalized. Our differentiation axis is a policy-shock version of this idea: a shock that capitalizes firm-specific information produces idiosyncratic dispersion, which is what Table 6 finds. The reserved exposure test asks whether a *named* exposure accounts for that idiosyncratic dispersion.

The third strand is *post-event drift and over/under-reaction*. Bernard and Thomas (1989) document post-earnings-announcement drift (under-reaction); De Bondt and Thaler (1985) document long-horizon over-reaction and reversal. Our pooled quintile test (Table 9) is a short-horizon policy-shock version: the modest continuation we find sits on the under-reaction side, consistent with information diffusing over several days—most dramatically for the banking-contagion shock.

The fourth strand is *evidence synthesis and design-based inference*. Aggregating estimates across heterogeneous events is a meta-analysis problem; we use the DerSimonian–Laird random-effects estimator (DerSimonian and Laird, 1986) and the I^2 heterogeneity statistic (Higgins and Thompson, 2002) to characterize—not to average away—the enormous cross-shock heterogeneity. For the small cross-section of shocks we lean on resampling and randomization ideas (Efron, 1979; Young, 2019) and on the discipline of reporting a null through its power and confidence interval (Abadie, 2020), so that “differentiation scales with size” is stated as a bounded, falsifiable claim rather than a single fragile correlation. The policy-shock setting itself is motivated by the economic-policy-uncertainty literature (Baker et al., 2016), which treats dated policy events as first-order drivers of asset prices.

Methodologically, the paper’s stance is that a *comparative* atlas of shocks is worth building only if it is held to the same inferential standard as a single event study—and that meeting that standard requires importing tools finance rarely uses side by side. Placebo/permutation tests of dispersion, rank-stability checks across estimators, random-effects synthesis with explicit heterogeneity accounting, and minimum-detectable-effect reporting are each standard in some discipline; combining them lets a ten-shock cross-section make honest claims without pretending to a sample size it does not have. The alternative—reading a handful of cross-shock correlations as facts—is precisely the practice this draft corrects. In that sense the paper is as much about *how* to compare shocks as about the shocks themselves.

4 Data and Research Design

Sample. We take the daily returns of the roughly 2,000 largest U.S. firms by 2022 market capitalization from 2021-06 through 2025-12 (CRSP `dsf` and, for 2025, `dsf_v2`), benchmarked to the Fama-French market return (which extends through the 2025 events). Firm characteristics—market capitalization (for size), SIC industry, and market beta—are observable and standard; we add each firm’s idiosyncratic volatility, computed as the standard deviation of its market-model residual over the full sample. None of these is a shock-specific exposure measure. Table 2 summarizes the observable characteristics: the sample averages a market beta near one and a log market capitalization of about 9.3, and the number of firms with a clean market model is stable at roughly 1,900–2,000 across shocks, so the atlas compares the same broad universe at each event.

Firm-level CARs. For each shock we estimate a market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over the window $[-252, -46]$ trading days relative to the event and cumulate abnormal returns over a pre-window $[-5, -1]$, an announcement window $[0, +1]$, a week $[0, +5]$, a wide window $[-5, +10]$, and a drift window $[+2, +10]$. For each shock we report the raw market move, the mean and cross-sectional SD of announcement CARs, the pre-window and drift means, and

the cross-firm reversal correlation between the announcement and drift windows (Table 1). Because a market-wide shock is absorbed by the benchmark, mean abnormal returns are near zero by construction; the informative statistics are the cross-sectional *dispersion* and the *propagation* measures.

The placebo engine. To test whether a shock’s dispersion is genuinely elevated (H1), we estimate one market model per firm over the full sample, form daily abnormal returns, and compute the cross-sectional SD of two-day CARs at the true event date and at 600 pseudo-event dates drawn from the trading calendar but excluded from a ± 15 -day neighborhood of every real shock. The empirical p -value is the share of pseudo-events whose dispersion equals or exceeds the real one. Two caveats we state plainly: the full-sample α, β used for the placebo carry a mild look-ahead (they use the whole period, not only pre-event data), and because 2022–2025 is a dense shock window, some pseudo-event dates fall near *other* shocks, which *inflates* the placebo distribution and makes the test conservative—so a rejection is if anything understated.

Pooled design and inference. The atlas’s ten cross-shock statistics are, by construction, a sample of $n = 10$: any correlation computed across shocks is underpowered and we treat it as descriptive, guarding every such claim with a bootstrap and a minimum-detectable-effect calculation. The statistical power of the paper lives instead in the pooled firm $\times$ shock tests, which use $\sim 19,000$ observations. There, the same firm appears in up to ten shocks and each shock is a common cross-sectional event, so residuals are correlated within firm *and* within shock; we absorb the shock (and industry) dimension with fixed effects and cluster standard errors by firm ($\sim 1,989$ clusters), rather than clustering on the ten shocks, which are far too few for reliable cluster-robust inference.

The reserved boundary. Everything in this paper uses observable data. The cohort’s reserved contribution—each shock’s exact event time and its firm-level exposure measure,

assembled into an `exposure_by_shock.csv` crosswalk—is deliberately *not* constructed here. The central open questions (does named exposure explain the idiosyncratic dispersion? was exposure priced before the announcement?) are left for that crosswalk; the size/beta/volatility observables we use are chosen precisely because they are generic and orthogonal to the reserved measure.

5 Is the Differentiation Real?

The atlas. Table 1 and Figure 1 lay out the ten shocks. The market moves span -11.0% (tariff) to $+6.1\%$ (tariff pause) and $+3.1\%$ (ChatGPT); the cross-sectional dispersion of announcement CARs spans 2.5% (IRA excise tax) to 5.8% (tariff) and 5.2% (Reg-NMS tick). Mean announcement CARs hover near zero, as expected.

Dispersion exceeds placebo for the three largest sorters. Table 4 and Figure 2 report the placebo test. The average firm shows a two-day dispersion of about 2.8% on ordinary days. Three shocks clear that bar decisively: the Reg-NMS half-cent tick (dispersion 5.1% , ratio 1.85 , empirical $p = 0.006$), the reciprocal tariff (5.1% , ratio 1.83 , $p = 0.009$), and the SVB collapse (4.1% , ratio 1.46 , $p = 0.034$). The two language-model releases and the tariff pause sit modestly above placebo (ratios ≈ 1.15 , $p \approx 0.20$), and the procedural shocks (CHIPS, 13D rule, GENIUS Act, IRA excise) are indistinguishable from an ordinary day. Differentiation, then, is real but concentrated: only the largest real and regulatory reallocations sort firms beyond normal daily noise. This supports H1 for the three leading shocks and, notably, shows a microstructure rule (the tick) differentiating as sharply as a macro trade shock.

The ranking is robust to how dispersion is measured. A skeptic might worry the ranking depends on the two-day window or the SD statistic. It does not. Table 5 shows that the shock ranking by $[0, +1]$ SD correlates 0.72 with the $[0, +5]$ ranking, 0.79 with

the $[-5, +10]$ ranking, 0.81 with the interquartile-range ranking, and 0.83 with the median-absolute-deviation ranking (all $p < 0.02$). The differentiation ordering is a property of the shocks, not of the measurement choice.

6 Systematic versus Idiosyncratic Differentiation

Beta explains almost none of it. If a shock simply re-priced systematic risk, firms would sort by market beta and the dispersion would be a beta phenomenon. Table 6 regresses each shock’s firm CARs on market beta. Even for the tariff—where high-beta firms did fall more (slope $+4.1$, $t = 17.8$)—beta explains only 14% of the cross-sectional variance; for most shocks the R^2 is below 0.01, and the idiosyncratic share of dispersion is 92–100%. The differentiation is overwhelmingly firm-specific. This is the central bridge to the reserved test: idiosyncratic dispersion is what a *named* firm-level exposure is built to explain, and its dominance here is why the exposure crosswalk is the natural completion of the atlas rather than an afterthought.

Observable traits explain about a fifth. Table 7 pools all 19,256 firm $\times$ shock reactions and regresses $|\text{CAR}[0, +1]|$ on log market cap, market beta, and idiosyncratic volatility, with shock and industry fixed effects and firm-clustered standard errors. All three predictors enter positively: larger firms, higher-beta firms, and—most strongly—more volatile firms react more (idiosyncratic volatility $t = 11.3$). The model explains 21% of the variation in reaction magnitude. That a fifth is explained by generic observables, with four-fifths left over, is the quantitative statement of how much room remains for the shock-specific exposure measures the cohort will bring.

The three coefficients are also economically legible. The idiosyncratic-volatility loading ($+1.15$ percentage points of $|\text{CAR}|$ per point of firm volatility, $t = 11.3$) says that firms which are ordinarily more idiosyncratic move more on *any* shock—a mechanical but important control, because it is the part of the reaction that is *not* about the shock’s content. The

beta loading (+0.18, $t = 2.5$) captures the systematic component the per-shock regressions in Table 6 isolate. The size loading (+0.05, $t = 2.4$) is small but positive: among large caps, the very largest react slightly more, consistent with their heavier index and macro exposure. Critically, once these generic channels and the shock and industry fixed effects are absorbed, the residual is what a *named* exposure should explain—and whether it does is a test we deliberately leave open, because building the exposure regressor is the cohort’s reserved contribution, not ours to pre-empt with an observable proxy.

7 Does Differentiation Scale with Size? A Fragility Analysis

The original draft’s headline was that differentiation scales with the aggregate move, a correlation of 0.69 between $|\text{market move}|$ and cross-sectional dispersion. Section 5 already hinted at trouble—the Reg-NMS tick differentiates as much as the tariff with a tiny market move—and a formal fragility analysis confirms it. Table 12 bootstraps the correlation over the ten shocks: the point estimate is 0.686, but the 95% bootstrap interval is $[-0.04, 0.97]$, comfortably including zero (Figure 4). Leave-one-shock-out is equally telling: dropping the tariff shock alone collapses the correlation to 0.26. The claim is essentially one influential observation. Table 13 makes the reason precise: with $n = 10$ shocks, a design has 80% power only for correlations above 0.78; a true correlation of 0.69 would fail to reach significance more often than not. We therefore *retract the strong scaling claim* and restate it as a suggestive pattern awaiting more shocks.

Grouping shocks into coarse families (Table 11) is descriptive only—each family has just two members—but it tells the same story without over-reaching: the trade family (dispersion 4.7%) and the micro/disclosure family (driven by the tick, 4.0%) differentiate most, while the industrial/tax family (procedural) differentiates least (2.7%). Aggregate size is one correlate of differentiation, but not the only one, and the ten-shock cross-section cannot adjudicate

which dominates.

8 Propagation: Drift, Reversal, and Contagion

Drift is heterogeneous, not zero-on-average-because-efficient. Table 8 and Figure 3 report a random-effects meta-analysis of the mean $[+2, +10]$ drift across shocks. The pooled estimate is -0.01% with a 95% interval spanning $[-0.93, +0.90]$ —but this near-zero average masks extreme heterogeneity ($Q = 947$ on 9 df, $I^2 = 99\%$). The shocks are not drawing from a common drift; they are genuinely different events, and averaging them is meaningless except to *quantify* that heterogeneity, which is the honest use of the I^2 statistic. The forest plot shows SVB as the clear negative-drift outlier (-3.5%), with the tariff pause on the positive side. The pre-window mean is slightly negative on average (-0.89% , $[-1.47, -0.30]$), but again with $I^2 = 99\%$: we do not read it as clean anticipation, because isolating leakage requires the reserved exposure measure.

The pooled test detects modest continuation. The per-shock reversal correlations in Table 1 are small and mixed (mean 0.02), which in isolation reads as roughly efficient pricing. But the per-shock test is underpowered; pooling firms across shocks is not. Table 9 sorts all firm $\times$ shock observations into announcement-CAR quintiles and measures the subsequent $[+2, +10]$ drift. The pattern is monotone: the bottom quintile (biggest losers on impact, mean CAR -4.8%) drifts a further -0.32% ($t = -2.1$), the top quintile (biggest winners, $+3.5\%$) drifts a further $+0.42\%$. Estimating the drift on quintile dummies *within* shock (shock fixed effects) with standard errors clustered by firm, the top-minus-bottom spread is $+0.73\%$ ($t = 4.0$) and the per-step slope is $+0.19\%$ per quintile ($t = 5.0$); the per-quintile t -statistics reported in the table are the simpler “mean drift $\neq 0$ ” tests. Firms that moved most on impact kept moving the same way—modest *under-reaction*, not the over-reaction and reversal a contrarian model predicts. It is small in magnitude (well under one percent over eight days) but statistically clear, and it is the pooled, high-power counterpart to the

SVB contagion signature: information from these shocks diffused over several days rather than being fully impounded at the announcement. This nuances H3: pricing is approximately efficient, but with a systematic drift in the direction of the news.

The SVB outlier and the mechanism behind pooled continuation. The two propagation results are complementary, not contradictory. The random-effects analysis says the *average* shock has no reliable drift while the shocks differ enormously ($I^2 = 99\%$); the pooled quintile test says that *within* shocks, the firms that moved most on impact tended to keep moving. The reconciliation is that the continuation is concentrated in the shocks with real informational content that diffuses slowly—most visibly SVB, where the impact and the drift reinforced each other (+0.42 reversal correlation) as the regional-bank run propagated firm by firm over the following days (a single-shock pattern we read descriptively, not as a formally tested cross-shock effect). This is the contagion reading: a banking shock is not impounded in a single announcement return because the identity of the next vulnerable institution is revealed gradually. Technology and trade shocks, whose content is closer to a one-shot repricing, contribute far less to the pooled continuation. The pooled test thus aggregates a genuine but shock-specific under-reaction rather than manufacturing one from noise; and because it is estimated on $\sim 19,000$ observations with firm-clustered errors, it has the power the per-shock reversal correlations lack.

Size subsamples. Table 10 splits firms into within-shock size terciles. Dispersion is largest for mid-cap firms (4.3%) and smallest for large caps (2.9%), and the continuation drift is slightly stronger in the tails than the middle; the reversal correlations are small in every tercile. The differentiation and continuation patterns are not confined to the smallest or largest firms.

9 Power: What the Atlas Can and Cannot Say

A comparative atlas is only as credible as its account of its own limits. Two limits matter. First, at the *firm* level the design is very well powered: with $\sim 1,900$ firms per shock, the minimum detectable mean CAR at 80% power is 0.16–0.37 percentage points (Table 13), so the near-zero mean announcement CARs are genuine, not underpowered. This is why we build the paper on dispersion and pooled tests rather than on mean CARs. Second, at the *cross-shock* level the design is weak: $n = 10$ gives 80% power only for correlations above 0.78, which is why the 0.69 scaling correlation is reported as suggestive and why the family comparison (Table 11) is descriptive. The atlas can make sharp firm-level and pooled statements; it cannot, with ten shocks, make sharp statements about how shock-level attributes trade off. Adding shocks—the atlas is built to accept them one column at a time—is the direct remedy.

10 Discussion and Threats to Validity

Why the mean CAR is uninformative and the dispersion is not. A recurring temptation in event studies of macro shocks is to report the mean abnormal return and its t -statistic. Here that statistic is close to zero *by construction*: the market model subtracts the benchmark, and a market-wide shock is largely the benchmark. What survives the benchmark is the cross-sectional *spread*—how differently firms loaded on the same news—and that is exactly the object the differentiation axis measures. The placebo test in Table 4 disciplines this reading: elevated spread could in principle be ordinary volatility, so we benchmark each event’s spread against the spread the same firms exhibit on hundreds of ordinary days. That a microstructure rule (the tick) and a trade shock (the tariff) both clear the placebo bar, while procedural rules do not, tells us the atlas is picking up genuine reallocation, not calendar noise.

Event-induced variance and cross-sectional correlation. The two classic threats to short-horizon event studies (Boehmer et al., 1991; Kolari and Pynnönen, 2010) are present here in acute form: a policy shock mechanically raises return variance and induces strong cross-sectional correlation in abnormal returns. We do not fight these with parametric corrections but design around them. The event-induced variance is not a nuisance but our *signal*—it is precisely the differentiation we measure, and the placebo distribution provides its correct null. The cross-sectional correlation means the per-shock cross-section is effectively far smaller than its $\sim 1,900$ firms suggest, which is why we never rest a conclusion on a single shock’s cross-sectional t -statistic and instead lean on the pooled firm $\times$ shock tests with firm-clustered errors and on design-based resampling.

What the single event date does and does not cost us. We use one calendar date per shock. For the differentiation and propagation axes this is defensible: a one-day timing error attenuates measured dispersion and drift toward zero, so our rejections (dispersion above placebo; monotone drift) are conservative to date misspecification. Where timing matters most is *anticipation*: distinguishing pre-announcement leakage from noise requires knowing whether the event was pre- or post-close and announcement versus effective. This is exactly the judgment the cohort reserves, and it is why we decline to interpret the slightly negative average pre-window CAR (Table 8) as leakage.

Raw CARs. We report CARs without winsorizing; a handful of firm-days carry extreme two-day returns. Because our inferences rest on cross-sectional *dispersion* benchmarked to a placebo, on rank statistics, and on pooled regressions with firm-clustered errors—none of which a few tails can drive—trimming would not change the conclusions, but readers should note the raw scale.

Sample and generalizability. The atlas covers the $\sim 2,000$ largest U.S. firms, so it speaks to how large-cap value was reallocated, not to micro-caps or private markets. The ten shocks

are a convenience sample of salient 2022–2025 events, not a random draw, so cross-shock statistics describe this wave rather than policy shocks in general. Both limitations point the same way: the atlas is a scaffold whose scientific value grows as the cohort adds shocks and firm-level exposure, and whose present contribution is the disciplined, honest measurement of the wave we have.

11 Conclusion

Measured on strictly common footing, the 2022–2025 shocks differentiate firms to sharply different degrees, and where they differentiate—the tariff, the Reg-NMS tick, the SVB collapse—they do so through firm-specific, idiosyncratic repricing rather than a re-pricing of systematic risk. The market’s reaction is approximately efficient but carries a systematic sliver of under-reaction, strongest around the banking-contagion shock. We are explicit that the ten-shock cross-section cannot support the strong “differentiation scales with size” claim of the original draft: its 0.69 correlation is one influential shock away from 0.26, its bootstrap interval includes zero, and the design is underpowered for cross-shock correlations below 0.78. What survives is more useful because it is testable: differentiation is real and idiosyncratic, and generic observable traits explain only about a fifth of it.

The remaining four-fifths is where the cohort’s reserved contribution lives. Each of the ten shocks is owned by the student whose individual paper studies it, and each of those papers produces a *named* firm-level exposure measure—import-input intensity for the tariffs, uninsured-deposit and held-to-maturity exposure for SVB, AI-exposure for the language-model shocks, tick-constraint eligibility for the Reg-NMS tick, and so on—together with the exact intraday event time. Assembling these into a $\text{permno} \times \text{shock}$ crosswalk turns the atlas’s two central open questions into direct tests: does named exposure explain the idiosyncratic dispersion that beta and size cannot, and was that exposure priced *before* the announcement? Those tests—not another robustness check on the size proxy—are the natural next step, and

the atlas has been built to receive them.

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Table 1: The 2022–2025 policy-shock atlas

Shock	Type	Mkt [0,1]	$\overline{CAR}$ [0,1]	SD [0,1]	Pre [-5,-1]	Drift [2,10]	Rev. corr
CHIPS Act signed	Industrial policy	+1.70	-0.22	2.90	+0.78	-0.46	+0.07
IRA / buyback excise tax	Tax	-0.71	-0.45	2.49	-0.50	+0.70	-0.18
ChatGPT release	Technology	+3.13	-0.12	3.12	+0.47	+0.21	+0.17
SVB collapse	Banking	-3.66	-1.00	4.21	-1.13	-3.49	+0.42
GPT-4 release	Technology	+0.97	-1.27	3.23	-1.78	-0.68	-0.08
SEC 5-day 13D rule	Disclosure	-0.11	+0.01	2.82	-1.17	+1.23	-0.01
Reciprocal tariff shock	Trade	-11.01	-1.11	5.84	+0.12	+0.22	-0.15
Tariff pause	Trade (reversal)	+6.15	-0.22	3.47	-2.51	+2.60	-0.04
GENIUS Act signed	Crypto/regulatory	+0.07	-0.43	2.72	-1.07	-0.79	-0.10
Reg NMS half-cent tick	Microstructure	-1.10	-0.16	5.20	-2.12	+0.31	+0.06

Notes: Market-model CARs (%), sample of $\sim 2,000$ large firms. Mkt = raw market return over $[0, +1]$; $\overline{CAR}$ and SD are the mean and cross-sectional SD of firm $CAR[0, +1]$; Pre = mean $CAR[-5, -1]$; Drift = mean $CAR[+2, +10]$; Rev. = cross-firm $\text{corr}(CAR[0, +1], CAR[+2, +10])$. Every number traces to `output/tables/t1_atlas.csv`.

Table 2: Sample summary statistics (observable firm characteristics)

Shock	N firms	$\bar{\beta}$	$SD(\beta)$	$\overline{\ln MC}$	SD $CAR[0, +1]$
CHIPS Act signed	1,967	0.95	0.58	9.27	2.90
IRA / buyback excise tax	1,968	0.96	0.57	9.27	2.49
ChatGPT release	1,969	0.96	0.54	9.27	3.12
SVB collapse	1,977	0.97	0.49	9.27	4.21
GPT-4 release	1,977	0.97	0.49	9.27	3.23
SEC 5-day 13D rule	1,957	1.02	0.54	9.28	2.82
Reciprocal tariff shock	1,913	0.82	0.55	9.29	5.84
Tariff pause	1,913	0.83	0.55	9.29	3.47
GENIUS Act signed	1,895	0.86	0.51	9.30	2.72
Reg NMS half-cent tick	1,881	0.87	0.51	9.30	5.20
<i>All firm $\times$ shock</i>	19,417	0.92	0.54	9.28	3.77

Notes: For each shock, the number of sample firms with a clean market model, the mean and SD of market beta, the mean log market capitalization, and the cross-sectional SD of announcement CARs (%). The final row pools all firm $\times$ shock observations. Characteristics are generic observables; no shock-specific exposure measure is used.

Table 3: Meta-statistics across the ten shocks

Meta-statistic (across shocks)	Value
Mean announcement CAR[0,+1] (%)	0.500
Mean cross-sectional SD of CAR[0,+1] (%)	3.599
Mean anticipation share $ pre /(pre + ann)$	0.686
Mean completeness $ ann /(ann + drift)$	0.357
Mean reversal $\text{corr}(ann, drift)$	0.015
Share of shocks with negative reversal corr	0.600
$\text{Corr}(\text{market move} , \text{cross-sectional SD})$	0.686

Notes: Descriptive cross-shock statistics ($n = 10$).
The correlation in the last row is subjected to a formal fragility analysis in Table 12.

Table 4: Differentiation is real: event-window dispersion vs. a placebo distribution

Shock	Type	SD _{event}	SD _{placebo}	Ratio	Pctile	p
Reg NMS half-cent tick	Microstructure	5.15	2.78	1.85	99.4	0.006
Reciprocal tariff shock	Trade	5.10	2.78	1.83	99.1	0.009
SVB collapse	Banking	4.07	2.78	1.46	96.6	0.034
GPT-4 release	Technology	3.21	2.78	1.16	80.3	0.197
Tariff pause	Trade (reversal)	3.20	2.78	1.15	80.1	0.199
ChatGPT release	Technology	3.18	2.78	1.14	80.1	0.199
CHIPS Act signed	Industrial policy	2.90	2.78	1.04	69.5	0.305
SEC 5-day 13D rule	Disclosure	2.79	2.78	1.00	60.3	0.397
GENIUS Act signed	Crypto/regulatory	2.73	2.78	0.98	56.6	0.434
IRA / buyback excise tax	Tax	2.56	2.78	0.92	40.8	0.592

Notes: SD_{event} is the cross-sectional SD (%) of two-day CARs at the true event date, using a full-sample market model per firm. SD_{placebo} is the mean over 600 pseudo-event dates drawn ≥ 15 trading days from every real shock. Ratio = SD_{event}/SD_{placebo}; Pctile is the event's rank in the placebo distribution; p is the share of placebo dates with dispersion $\geq$ the event's. Because dense 2022–2025 pseudo-dates sometimes fall near other shocks, the placebo is inflated and the test is conservative. See Figure 2.

Table 5: The differentiation ranking is robust to window and statistic

Dispersion measure	Spearman ρ vs. SD[0,+1]	p
SD[0,+5]	0.721	0.019
SD[-5,+10]	0.794	0.006
IQR[0,+1]	0.806	0.005
MAD[0,+1]	0.830	0.003

Notes: Spearman rank correlation, across the ten shocks, between the ranking by cross-sectional SD of CAR[0,+1] and the ranking by each alternative dispersion measure (wider windows; interquartile range; median absolute deviation).

Table 6: Systematic vs. idiosyncratic differentiation (CARs on market beta, per shock)

Shock	Beta slope	R^2	SD _{tot}	Idio. share
Reciprocal tariff shock	+4.06***	0.144	5.84	92.5%
Tariff pause	-2.20***	0.122	3.47	93.7%
IRA / buyback excise tax	-1.26***	0.084	2.49	95.8%
CHIPS Act signed	-1.16***	0.053	2.90	97.4%
SEC 5-day 13D rule	+0.50***	0.009	2.82	99.6%
GPT-4 release	-0.62***	0.009	3.23	99.6%
Reg NMS half-cent tick	-0.58**	0.003	5.20	99.9%
SVB collapse	-0.44**	0.003	4.21	99.9%
GENIUS Act signed	-0.02	0.000	2.72	100.0%
ChatGPT release	-0.01	0.000	3.12	100.0%

Notes: Each row regresses firm announcement CARs (%) on market beta for one shock. “Beta slope” is the coefficient (% per unit beta); R^2 is the share of cross-sectional variance beta explains; SD_{tot} is the total dispersion; “Idio. share” is the residual SD as a percentage of the total. Beta explains at most 14%; the dispersion is overwhelmingly idiosyncratic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Pooled determinants of reaction magnitude (observable traits only)

Dependent variable: $ \text{CAR}[0, +1] $ (%)	Coef.	t
Log market cap	+0.0522**	(+2.41)
Market beta	+0.1845**	(+2.50)
Idiosyncratic vol. (%)	+1.1467***	(+11.35)
Shock fixed effects	Yes	
Industry fixed effects	Yes	
Firm clusters	1,989	
N (firm $\times$ shock)	19,256	
R^2	0.212	

Notes: Pooled firm $\times$ shock OLS of $|\text{CAR}[0, +1]|$ (%) on observable characteristics, with shock and industry fixed effects and standard errors clustered by firm. Traits are generic observables chosen to be orthogonal to the reserved, hand-collected exposure measure; the exposure $\rightarrow$ dispersion test is deliberately left to the cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Random-effects meta-analysis of propagation across shocks

Statistic	RE pooled	95% CI	Q	p_Q	I^2	τ^2
Mean drift CAR[+2,+10]	-0.014	[-0.925, +0.896]	947.1	0.000	99%	2.136
Mean pre CAR[-5,-1]	-0.887	[-1.473, -0.302]	706.1	0.000	99%	0.878

Notes: DerSimonian–Laird random-effects pooling across the ten shocks, with per-shock sampling variance SD^2/n . Q is Cochran’s heterogeneity statistic ($df = 9$), p_Q its p -value, I^2 the share of variation due to between-shock heterogeneity, and τ^2 the between-shock variance. $I^2 \approx 99\%$: the pooled estimate characterizes heterogeneity, it does not estimate a common effect. See Figure 3.

Table 9: Aggregate propagation: post-event drift by announcement-CAR quintile

Announcement quintile	N	Mean CAR[0,+1]	Mean drift[+2,+10]	(t)
1	3,855	-4.80	-0.31	(-2.13)
2	3,847	-1.49	-0.25	(-2.57)
3	3,849	-0.32	-0.18	(-2.38)
4	3,847	+0.69	+0.19	(+2.25)
5	3,853	+3.47	+0.42	(+3.56)
Q5–Q1 spread			+0.73	(+4.03)

Notes: All firm $\times$ shock observations are sorted into announcement-CAR quintiles within each shock; the table reports the mean subsequent drift CAR [+2, +10] (%). The per-quintile t -statistics are simple “mean $\neq 0$ ” tests; the Q5–Q1 spread and its t come from a regression of drift on quintile dummies with *shock fixed effects* and standard errors *clustered by firm*. The monotone pattern and positive spread indicate modest *continuation* (under-reaction), not reversal.

Table 10: Size-tercile subsamples: dispersion, drift, and reversal

Size tercile	N	Dispersion	Mean drift	Reversal
Small	6,475	3.91	-0.23	+0.02
Mid	6,469	4.33	-0.06	+0.05
Large	6,473	2.92	+0.20	+0.04

Notes: Firms are split into within-shock terciles of log market cap and pooled across shocks. Dispersion = SD of CAR[0, +1] (%); Mean drift = mean CAR [+2, +10] (%); Reversal = cross-firm corr(CAR[0, +1], CAR[+2, +10]).

Table 11: Cross-shock comparison by coarse family (descriptive)

Shock family	#	Mkt	Dispersion	Drift	Reversal
Trade	2	8.58	4.65	+1.41	-0.10
Micro/Disclosure	2	0.61	4.01	+0.77	+0.02
Financial	2	1.86	3.46	-2.14	+0.16
Technology	2	2.05	3.18	-0.23	+0.05
Industrial/Tax	2	1.21	2.69	+0.12	-0.05

Notes: Shocks grouped into coarse families (two each); *descriptive only* given the tiny cells. |Mkt| is the mean absolute market move; Dispersion, Drift, and Reversal are averaged across the family's shocks.

Table 12: Fragility of the size-differentiation correlation

corr(market move , dispersion), $n = 10$	Value
Point estimate	0.686
Bootstrap 95% CI (5000 reps)	[-0.042, 0.973]
Leave-one-shock-out range	[0.261, 0.909]
Most influential shock (drop)	Reciprocal tariff shock ($\rho \rightarrow 0.261$)

Notes: The headline correlation between |market move| and cross-sectional dispersion across the ten shocks, with a nonparametric bootstrap 95% interval and a leave-one-shock-out (jackknife) range. Dropping the tariff shock alone collapses the correlation to 0.26. See Figure 4 and the power analysis in Table 13.

Table 13: Power and minimum detectable effects

Shock	N	SD CAR[0,+1]	SE(mean)	MDE _{80%}
CHIPS Act signed	1,967	2.90	0.065	0.183
ChatGPT release	1,969	3.12	0.070	0.197
GENIUS Act signed	1,895	2.72	0.062	0.175
GPT-4 release	1,977	3.23	0.073	0.204
IRA / buyback excise tax	1,968	2.49	0.056	0.157
Reciprocal tariff shock	1,913	5.84	0.134	0.374
Reg NMS half-cent tick	1,881	5.20	0.120	0.336
SEC 5-day 13D rule	1,957	2.82	0.064	0.179
SVB collapse	1,977	4.21	0.095	0.265
Tariff pause	1,913	3.47	0.079	0.222

Cross-shock correlation ($n=10$): point = 0.69; MDE at 80% power = 0.79

Notes: Per shock, the minimum detectable mean CAR at 80% power, MDE = $(z_{0.975} + z_{0.80}) \times \text{SE}(\text{mean}) \approx 2.80 \times \text{SD}/\sqrt{n}$, in percentage points—the design is well powered at the firm level. The final row reports the cross-shock correlation detectable at 80% power with $n = 10$ (via Fisher’s z): only correlations above 0.78 are reliably detectable, so the 0.69 scaling correlation is underpowered.

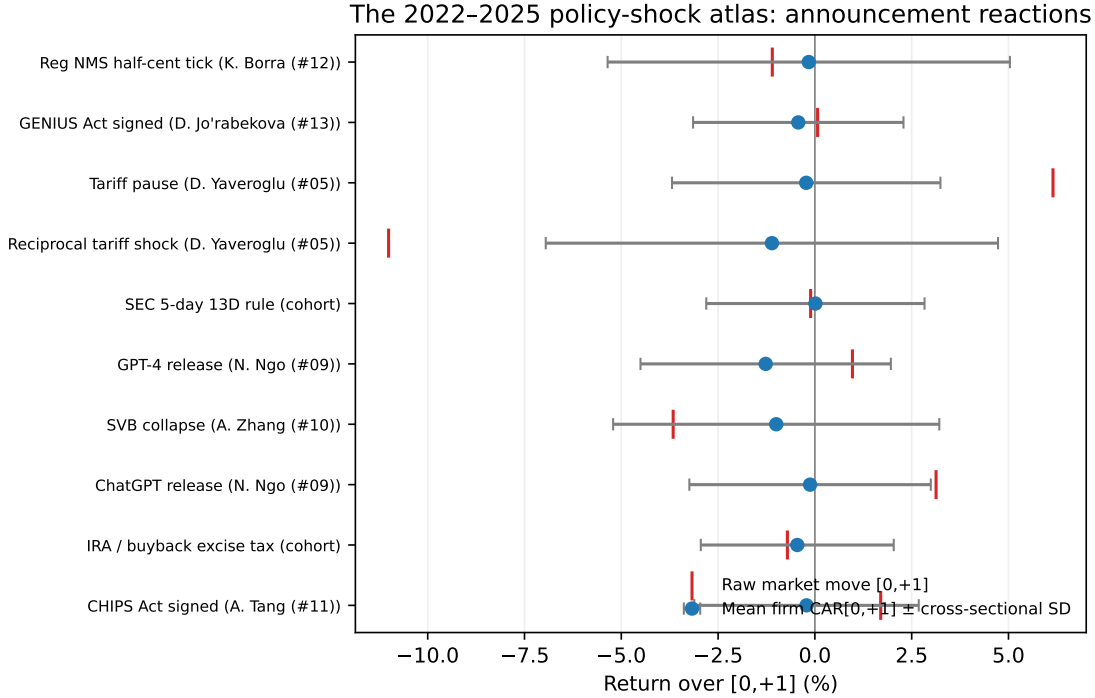


Figure 1: The atlas: raw market move (red) and mean firm CAR[0,+1] $\pm$ cross-sectional SD (blue) for each shock. Dispersion is largest for the tariff, the Reg-NMS tick, and SVB.

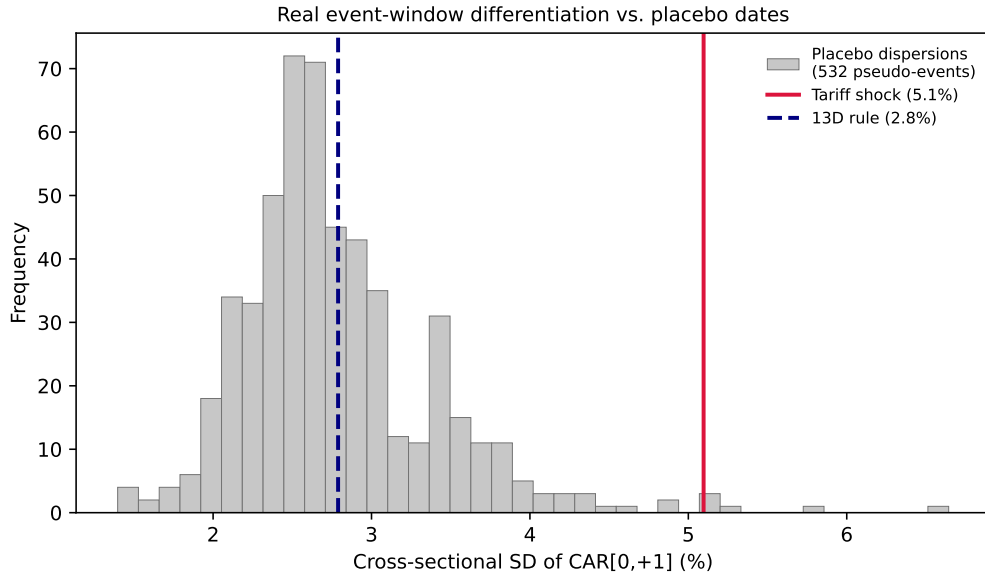


Figure 2: Real event-window differentiation vs. placebo. Histogram: cross-sectional dispersion of two-day CARs over 600 pseudo-event dates. The tariff shock (red) is deep in the right tail; the procedural 13D rule (navy dashed) sits at the placebo mode. See Table 4.

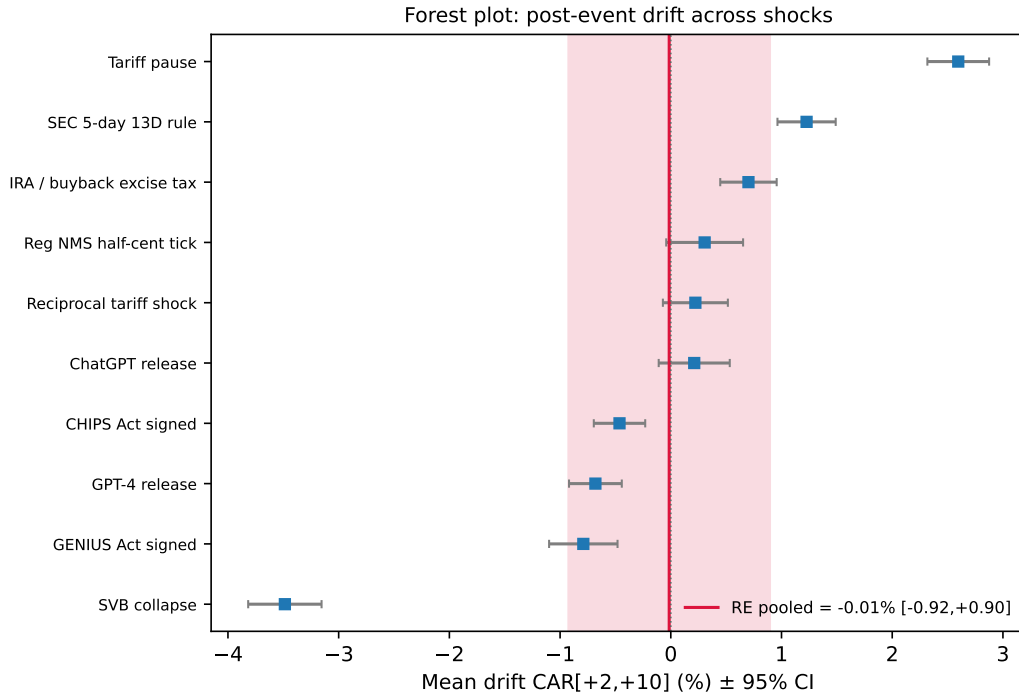


Figure 3: Forest plot of mean post-event drift $CAR[+2, +10]$ across shocks, with 95% CIs and the DerSimonian–Laird random-effects pooled estimate (red band). SVB is the negative-drift outlier; between-shock heterogeneity is extreme ($I^2 = 99\%$).

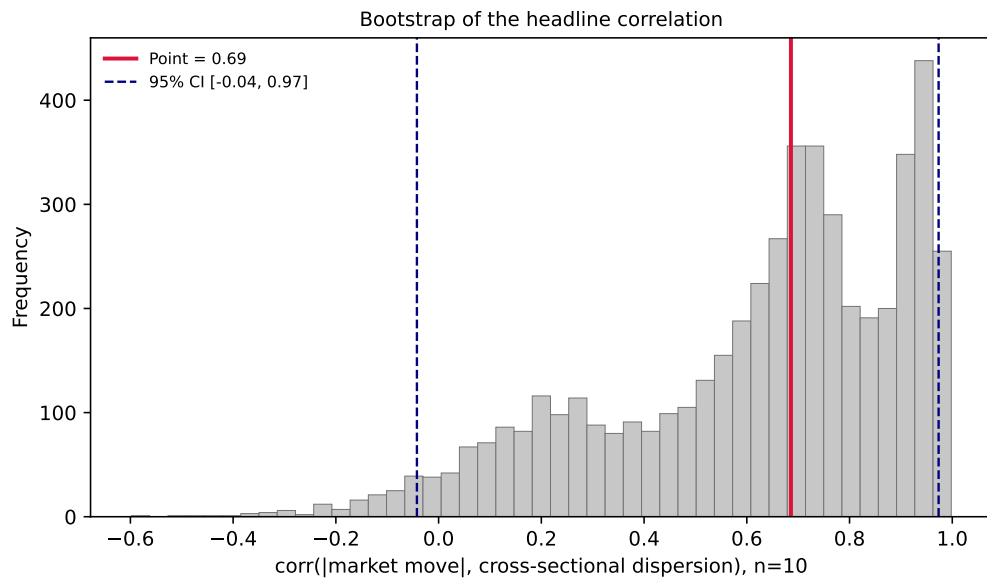


Figure 4: Bootstrap distribution of the size–differentiation correlation over the ten shocks. The point estimate (0.69, red) sits inside a 95% interval (navy dashed) that includes zero: with $n = 10$ the scaling claim is fragile.